

How fast does Normal happen?

Shayaan Tanveer · Math 167R · R + Quarto

Question 1 — Sample size: Is $n = 30$ a universal rule? No. I investigated practical sample-size boundaries across five required populations and three extra-credit processes. The updated student guidance defines this scope; older lesson activities are background, not additional required populations.

My decisions, and the argument behind them

Case	n	Why I accept it	Why not smaller?
Normal	1	Exact Normality at every n.	No positive n below 1.
Six-sided die	10	Symmetric bell shape; modest short-tail error and fine steps.	5 has clearer short-tail error; 6-9 are borderline.
Exponential	90	Good central shape; modest right skew, correct SE.	30-60 bend more; 75-80 are borderline across seeds.
Binomial small	75	Modest skew; mostly lattice steps, correct SE.	20-50 show stronger steps/skew; 60 is borderline.
Cauchy	None	No finite n: averages remain Cauchy.	Larger n cannot fix missing mean and variance.
Dependent failure	350	Bell-shaped empirical means; modest skew. No iid SE claim.	100-200 curve more; 300 is borderline. CLT unproved.
Basketball shots	140	Small skew; fine steps and correct varying-p SE.	30-100 have larger gaps; 110-125 are borderline.
Synthetic ages	15	Irregular mixture smooths quickly; modest left skew.	5-10 are more asymmetric; 12 is borderline.

My standard is practical, not exact. I accept small tail bends and discrete steps when the histogram, Q-Q plot, shape diagnostics, valid theory and nearby larger n jointly support the approximation. This is not a guarantee for extreme-tail probabilities.

Boundary uncertainty: these are judgment calls, not proved transition integers. A borderline smaller n may also be defensible under a slightly looser tolerance. I do not claim a nicer larger plot proves that its neighbor fails.

02 / THE METHOD

Question 2 — How do I decide Normal enough?

n counts observations inside one sample. $B = 10,000$ counts separate simulated samples. More B makes the simulation more precise; more n changes the distribution of the mean.

I started broad, then refined separately for each population. I compare histograms, Q-Q plots, skewness, excess kurtosis, mean and SE. Q-Q RMSE compares sorted, empirically standardized means with standard-Normal quantiles. It is a diagnostic, not a pass/fail test.

Calibration without a fake cutoff. Thirty repeated Normal experiments at each coarse n show ordinary Monte Carlo noise. At $n = 1$, the calibration 2.5th-97.5th percentile range for Q-Q RMSE is 0.010-0.018. Other populations need not match that noise to be useful approximations.

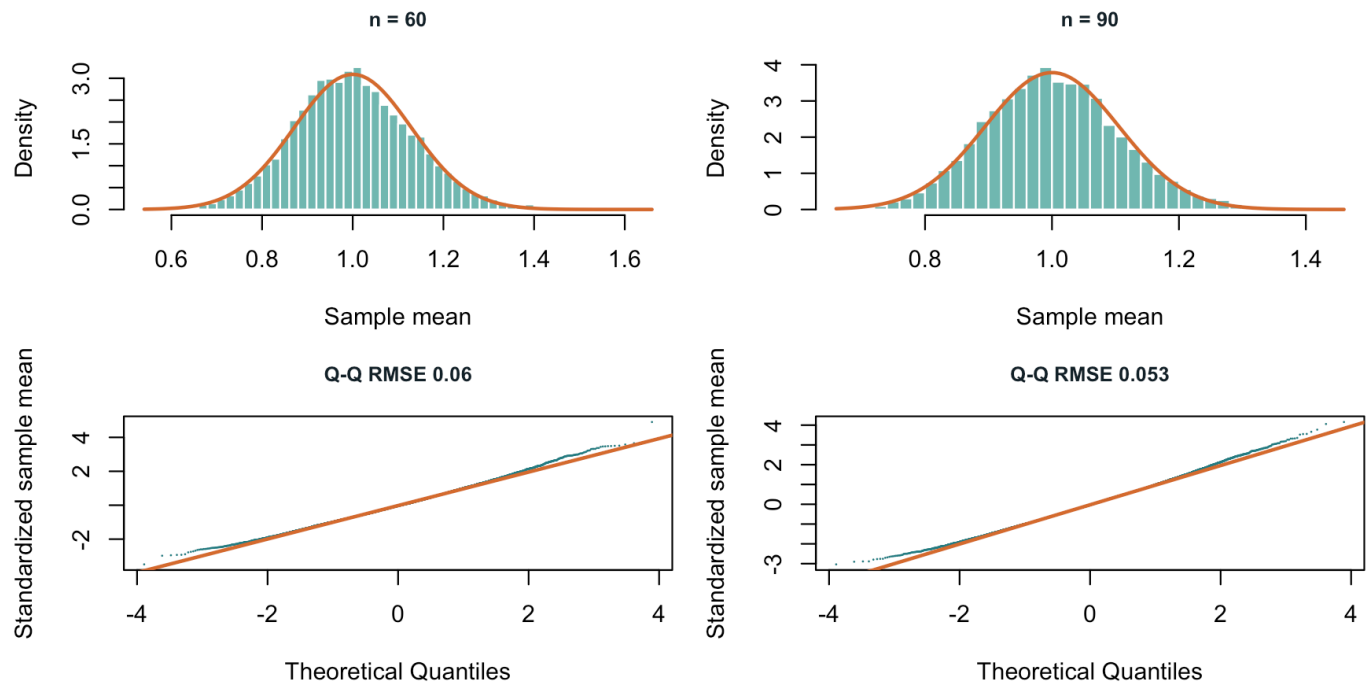


Figure 1: Exponential boundary evidence. Seed 7616, $B = 10,000$. Orange histogram overlays use mean 1 and $SE\ 1/\sqrt{n}$. Residual Q-Q curvature is visible even at the accepted n .

Why 90 rather than 60? The tail bend and skew are smaller, both seeds support similar discrepancy, and larger nearby values remain comparable. The theoretical skewness at 90 is 0.211, not zero. Values 75-80 are borderline, not proven failures.

03 / THE EXCEPTIONS

Question 3 — What if assumptions fail?

Cauchy: increasing n cannot rescue the mean

Independent standard Cauchy averages remain standard Cauchy at every n . Population mean and variance do not exist; $\sigma/\sqrt{n}$ is invalid. Finite simulated means and SDs are calculable, but do not estimate existing population moments. Testing through 5,000 illustrates this theoretical failure.

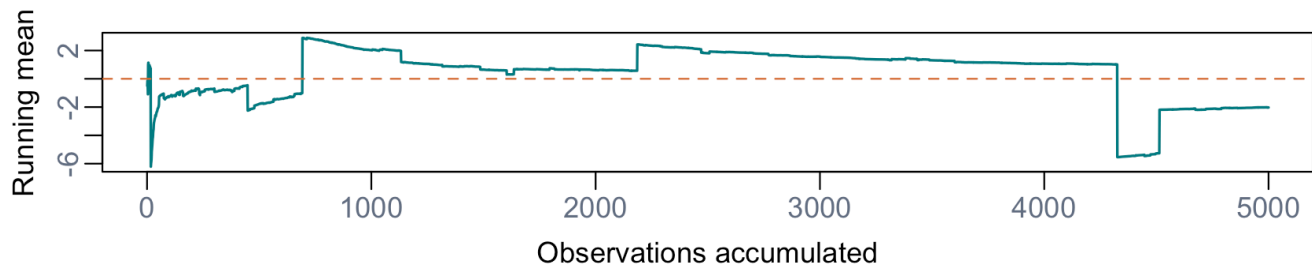


Figure 2: One Cauchy sequence, seed 7616. A running mean remains sensitive to extreme observations; the zero line marks location, not a population mean.

Dependence: a qualified empirical approximation

The next failure rate is $\exp(1.25 \times (\text{previous time} - 1))$. I preserve that recurrence within every sample. A long time tends to be followed by a shorter time. Covariance enters $\text{Var}(\bar{X})$, so I report the **empirical SE**, not an iid theoretical SE.

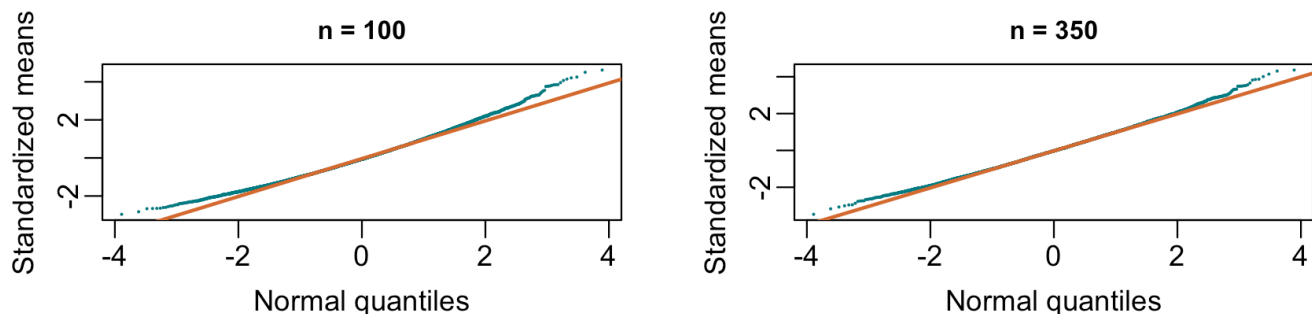


Figure 3: Dependent failure, seed 7616. At 350 the central Q-Q shape is closer to linear, but residual right skew remains. This is not a proof of a dependent CLT.

Individual times have bounded second moments because rates are bounded below. That does **not** establish a dependent CLT. My choice of 350 is empirical and qualified; exact boundary and asymptotic behavior remain unresolved. This differs fundamentally from Cauchy's proved non-convergence to Normal.

04 / THE CHECKS

The numbers behind the choices.

All values below come from generated diagnostic rows at the selected n . **Kurtosis means excess kurtosis.** Q-Q values use seeds 7616 and 7617; N/A indicates an invalid or unavailable quantity, not zero.

Case	n	Skew	Kurtosis	QQ 7616	QQ 7617	SE sim.	SE theory
Normal	1	0.003	-0.022	0.011	0.011	1.001	1.000
Six-sided die	10	0.000	-0.146	0.057	0.058	0.539	0.540
Exponential	90	0.214	0.032	0.053	0.054	0.106	0.105
Binomial small	75	0.109	0.050	0.057	0.060	0.077	0.077
Cauchy	None	N/A	N/A	N/A	N/A	N/A	N/A
Dependent failure	350	0.208	0.097	0.051	0.049	0.115	N/A
Basketball shots	140	-0.052	-0.039	0.057	0.061	0.038	0.037
Synthetic ages	15	-0.191	-0.039	0.050	0.056	7.128	7.135

Seed sensitivity: exact diagnostics change, but the selected values remain practical candidates under both seeds. The exact minimum is not immutable. I use one fixed alternate seed, never whichever seed produces the nicest picture. Saved vectors and CSV rows are cross-checked before plotting.

What mattered beyond the ordinary iid CLT

Basketball: independent shots have n probabilities spanning .90 to .50 ($n = 1$ uses .90). Expected mean is their average; variance is $\sum[p_i(1-p_i)]/n^2$. The iid $\bar{p}$ shortcut overstates variance. Independence is preserved; identical distribution is not needed for this bounded independent case.

Synthetic ages: the instructor's irregular mixture on 0-100 has mean 49.981 and SD 27.635. It is not historical mortality data. A complex-looking population need not be the slowest to produce a useful Normal approximation.

AI iteration, and my responsibility

I prompted AI to build and revise the R project. The revision fixed overwritten seed outputs, inappropriate shared refinement grids and overly strict benchmark matching. I questioned whether choosing a nicer plot answered the smallest- n question; that led to smaller-candidate checks and explicit boundary uncertainty. I approved a practical standard, not exact tail accuracy.

My role is prompting, checking the evidence and understanding the important code. The interaction log records actual exchanges; it does not claim a personal pre-run prediction or complete line-by-line mastery. Each separate investigation includes theory, plots, diagnostics, decision reasoning and code explanations.

This PDF is the complete reading submission. The following pages contain all eight investigations, numerical evidence, calibration, code explanations, actual AI iterations and reproduction details. The repository is supplementary, not required to follow the argument.

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Fastest: Normal is exact at $n = 1$. Among non-Normal iid cases, the symmetric die (10) and bounded synthetic mixture (15) were the fastest practical choices. **Slowest:** the dependent process (350, qualified), basketball shots (140), Exponential (90) and small Binomial (75) needed more observations under my standard. Cauchy is not part of this race: no finite n gives Normal convergence.

The comparison is gradual, not a switch at one integer. More skew, tail discrepancy or visible lattice steps can demand more n . Population complexity by itself is not a convergence rate: the two-peaked age mixture became useful much sooner than the simple Exponential.

What $n = 30$ does here: it is unnecessary for exact Normality and more than my practical die/age choices; it leaves clearer departures in the Exponential, small Binomial, basketball and dependent cases; it cannot repair Cauchy. These conclusions concern a central-distribution approximation, not certified accuracy for rare tail events.

Method, calibration and assumptions

I generate B independent Monte Carlo samples from each specified population/process and take one mean per sample. This is **not a bootstrap** from an observed dataset. The original population does not become Normal; the distribution of its averages may. The B means form the empirical sampling distribution.

The broad grid is 1, 2, 3, 5, 10, 15, 20, 30, 50, 75, 100, 150, 200, 300, 500. Each case adds its own refinement and large- n experiments. All tested n and their actual diagnostics appear below. Some older refinement points are retained as evidence, not treated as automatic selections. I inspect smaller candidates and nearby larger values, then make a judgment; no threshold algorithm chooses n .

What each diagnostic tells me

- **Histogram + valid Normal overlay:** symmetry, bell shape, center and spread. Orange overlays use theoretical mean/SE only when available. A bin choice can hide or exaggerate departures, so I also use Q-Q plots.
- **Normal Q-Q plot:** systematic curvature, extreme tails and lattice steps. Orange is the fitted quartile reference line, not a theoretical proof. The plotted means are standardized using their empirical mean and SD.
- **Skewness / excess kurtosis:** asymmetry and fourth-moment tail discrepancy, respectively; Normal reference values are zero. Finite- B estimates fluctuate.
- **Mean and SE:** empirical mean is the average of B means; empirical SE is their SD. Correct center/spread does not imply the entire shape is Normal.
- **Relative SE error:** absolute empirical-minus-theoretical SE divided by theoretical SE. N/A means no valid/available reference.
- **Q-Q RMSE:** square root of the average squared distance between sorted standardized means and $qnorm(ppoints(B))$. Smaller is closer on this diagnostic, not a universal acceptance certificate. No normality hypothesis-test p -value is used.

Calibration: genuinely Normal experiments still fluctuate

Thirty independent Monte Carlo experiments at each coarse n (each $B = 10,000$) provide the following empirical 2.5th–97.5th percentile ranges. With only 30 calibration experiments these ranges are approximate simulation context, not exact confidence limits or pass/fail cutoffs.

n	Skewness	Excess kurtosis	Q-Q RMSE
1	–0.033 to 0.032	–0.070 to 0.106	0.010 to 0.018
10	–0.052 to 0.049	–0.063 to 0.105	0.010 to 0.021
30	–0.053 to 0.030	–0.114 to 0.089	0.010 to 0.020
100	–0.041 to 0.046	–0.063 to 0.070	0.011 to 0.016
500	–0.041 to 0.029	–0.093 to 0.076	0.009 to 0.018

My accepted non-Normal cases generally remain outside Normal-only simulation noise. I explicitly tolerate modest residual curvature/steps for a practical central approximation. Demanding indistinguishability from exact Normality would answer a stricter question and give much larger n .

Which CLT assumptions matter?

For iid observations with finite mean μ and variance σ^2 , the standardized mean approaches $N(0,1)$; $E(\bar{X}) = \mu$ and $SE(\bar{X}) = \sigma/\sqrt{n}$. Quadrupling n halves that SE, but does not automatically remove all shape discrepancy. Increasing B cannot fix an invalid CLT assumption.

Case	Independent	Identical	Finite variance	Ordinary iid SE
Normal	TRUE	TRUE	TRUE	Valid
Six-sided die	TRUE	TRUE	TRUE	Valid
Exponential	TRUE	TRUE	TRUE	Valid
Binomial small	TRUE	TRUE	TRUE	Valid
Cauchy	TRUE	TRUE	FALSE	Invalid
Dependent failure	FALSE	FALSE	TRUE	Not used
Basketball shots	TRUE	FALSE	TRUE	Use p_i formula
Synthetic ages	TRUE	TRUE	TRUE	Valid

Heavy tails: Cauchy fails the finite-moment condition. Its average remains Cauchy exactly, not merely “too slow in my tested range.” **Dependence:** machine times have bounded individual second moments, but covariance changes the variance of the average. I have not proved a dependent CLT. **Non-identical observations:** independent bounded basketball outcomes can still have an approximately Normal average; the expected mean and variance must reflect every p_i . Identical distribution is sufficient for the ordinary theorem, not necessary for all CLTs.

The dependent-process variance identity is

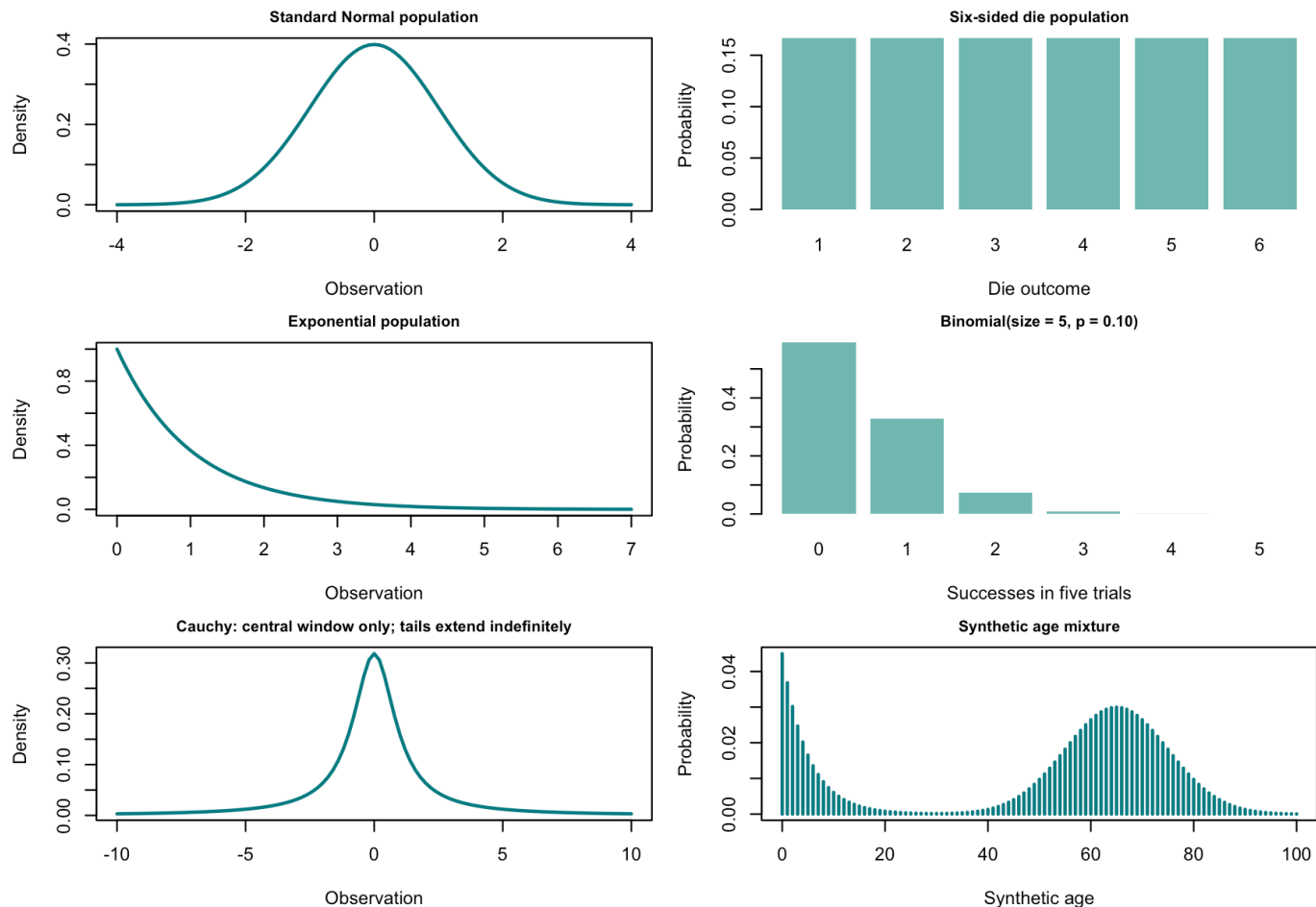
$$\text{Var}(\bar{X}) = \frac{1}{n^2} \left[\sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \right].$$

Covariances can reduce or increase variance; dependence does not always inflate SE. The supplied recurrence produces a compensating pattern, so blindly using baseline Exponential SD = 1 would be unjustified.

Important interpretation limits: a simulated histogram is evidence, not a theorem; exact minima are subjective transition regions; two seeds assess reproducibility but not every possible seed. No historical pre-run prediction was saved. The expectations below are labeled retrospective theory-based expectations, not invented memories.

Population shapes before averaging

These are exact model densities/probabilities, not simulated means. Normal, Exponential and Cauchy are continuous; die, Binomial and synthetic ages are discrete. Continuous curves show only the labeled central plotting ranges; tails do not end there. Basketball and machine-process constructions are visualized later because there is no single identical observation law across all indices.



These differing original shapes make the mean-distribution comparisons meaningful. A bell-shaped Cauchy center cannot reveal its infinite moments; theory and tail evidence matter. Discrete Q-Q steps in the mean plots are expected rather than programming errors.

01 / Normal

Required investigation. Selected n: 1. Exact by theory.

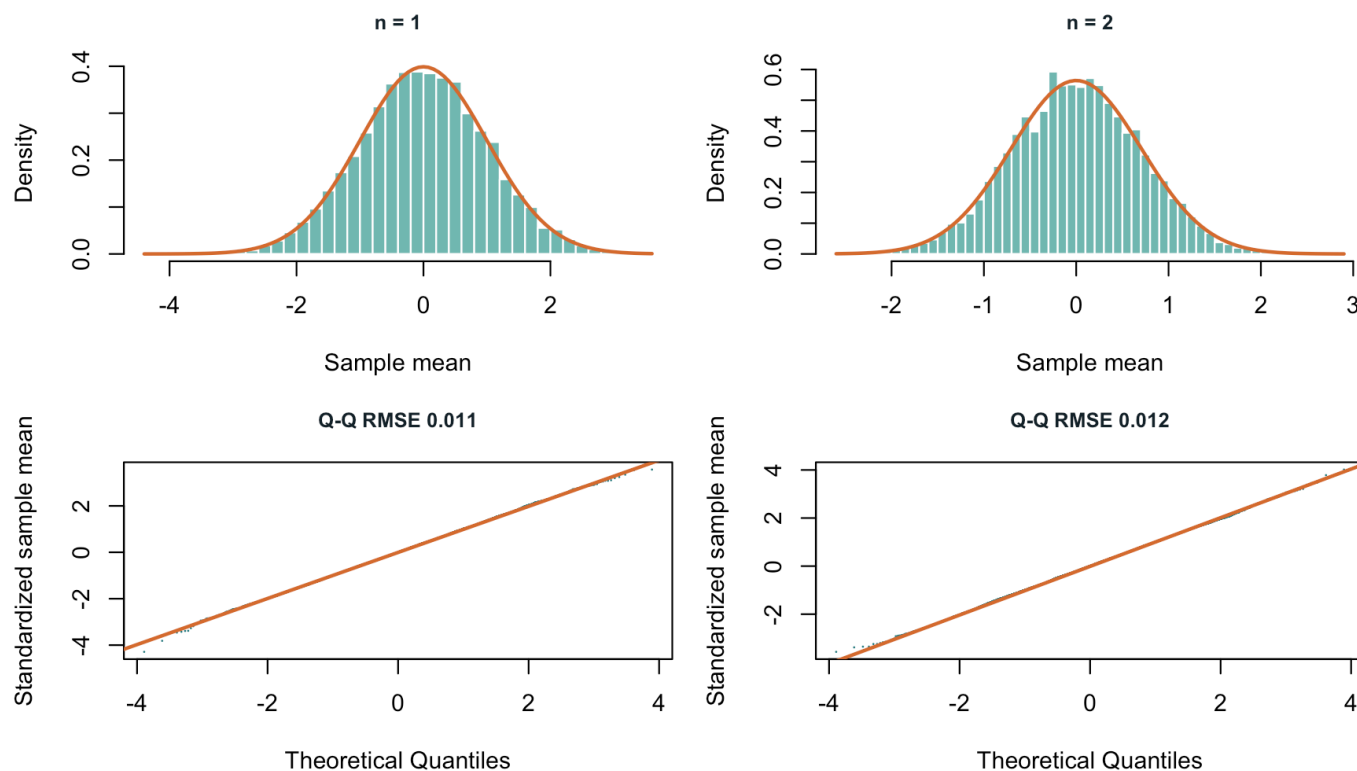
Population, assumptions and theory

Independent $N(0,1)$ observations: symmetric, unbounded and bell-shaped. Population mean 0, SD 1. The mean is exactly $N(0,1/n)$ for every positive n , not just asymptotically.

Retrospective theory-based expectation: Exact Normality at $n=1$ should validate the generator, plots and SE; larger n only shrinks spread.

Boundary evidence

Fresh samples, primary seed 7616, $B = 10,000$ at each n . Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: Exactly Normal for every positive n ; simulated shape and SE validate the implementation.

Why I did not choose smaller tested n: No smaller positive sample size exists.

Issue / noteworthy finding: Exact Normality is theoretical, not a Monte Carlo discovery.

Normal / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	-0.001	1.001	1.000	0.123	0.003	-0.022	0.011
2	0.000	0.704	0.707	0.413	-0.013	0.005	0.012
3	-0.001	0.575	0.577	0.484	-0.033	-0.041	0.016
5	0.002	0.449	0.447	0.494	-0.002	-0.065	0.012
10	0.000	0.318	0.316	0.676	0.004	-0.011	0.015
15	-0.002	0.260	0.258	0.881	-0.034	-0.055	0.013
20	-0.003	0.224	0.224	0.032	0.017	0.087	0.014
30	-0.003	0.183	0.183	0.361	-0.027	0.117	0.018
50	0.001	0.141	0.141	0.341	0.058	-0.059	0.018
75	0.000	0.116	0.115	0.719	0.029	0.043	0.013
100	0.000	0.099	0.100	0.743	0.001	0.033	0.009
150	-0.001	0.082	0.082	0.130	0.052	-0.073	0.017
200	0.000	0.070	0.071	0.329	-0.006	0.021	0.009
300	0.000	0.057	0.058	0.914	-0.005	-0.071	0.012
500	0.000	0.045	0.045	0.647	-0.011	0.010	0.016

Theoretical expected mean: 0

At the selected n , empirical mean = -0.00133 ; expected mean = 0.00000 . Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
1	0.003	-0.007	-0.022	0.007	0.011	0.011
2	-0.013	-0.030	0.005	-0.004	0.012	0.016

02 / Six-sided die

Required investigation. Selected n: 10. Practical judgment; borderline neighbors.

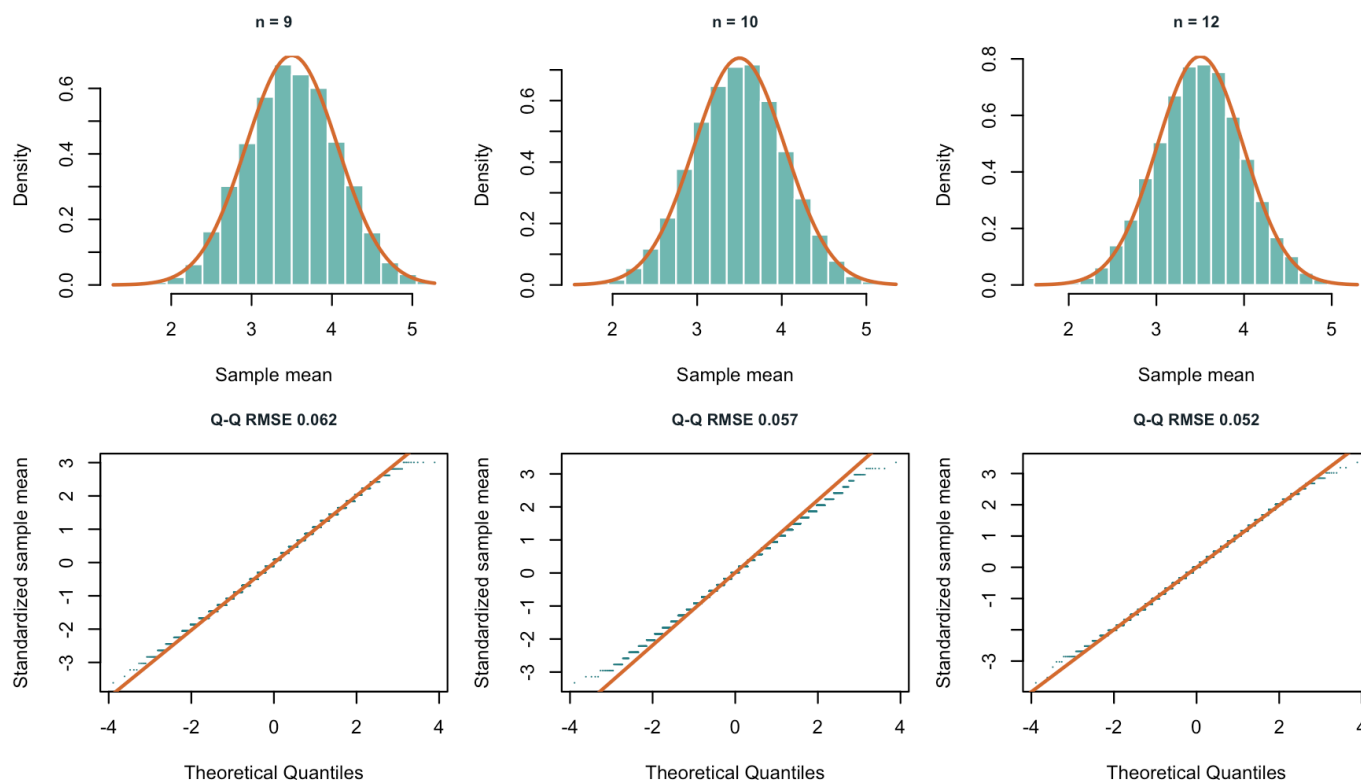
Population, assumptions and theory

Independent equally likely outcomes 1 through 6. Population mean 3.5; variance $35/12$; SD $\sqrt{35/12}$. The population is flat and discrete, while averages remain on a finer $1/n$ lattice.

Retrospective theory-based expectation: Symmetry and bounded observations should give relatively fast convergence, although discreteness and short tails remain visible.

Boundary evidence

Fresh samples, primary seed 7616, $B = 10,000$ at each n . Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: Symmetric bell shape, modest short-tail discrepancy and correct SE; both seeds give Q-Q RMSE about 0.057.

Why I did not choose smaller tested n: At $n=5$ the short-tail mismatch is clearer; 6-9 are borderline with coarser steps. I use 10 as the first convincing compromise, not proof that 9 fails.

Issue / noteworthy finding: Discrete means can be approximately Normal without losing Q-Q steps.

Six-sided die / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	3.518	1.710	1.708	0.104	-0.012	-1.269	0.310
2	3.506	1.214	1.208	0.492	0.001	-0.644	0.159
3	3.512	0.987	0.986	0.061	-0.006	-0.465	0.116
5	3.505	0.762	0.764	0.246	0.001	-0.236	0.081
6	3.496	0.700	0.697	0.363	0.006	-0.206	0.073
7	3.498	0.645	0.645	0.114	0.016	-0.128	0.066
8	3.504	0.611	0.604	1.132	0.032	-0.236	0.066
9	3.507	0.571	0.569	0.253	0.013	-0.197	0.062
10	3.495	0.539	0.540	0.121	0.000	-0.146	0.057
12	3.501	0.496	0.493	0.671	0.010	-0.154	0.052
15	3.500	0.439	0.441	0.344	-0.009	-0.117	0.046
18	3.501	0.405	0.403	0.697	0.002	-0.071	0.042
19	3.501	0.392	0.392	0.065	0.021	-0.128	0.043
20	3.498	0.383	0.382	0.325	-0.034	-0.100	0.041
21	3.499	0.374	0.373	0.234	-0.016	-0.143	0.042
25	3.498	0.341	0.342	0.048	0.025	-0.061	0.038
30	3.498	0.313	0.312	0.426	-0.024	-0.129	0.036
50	3.498	0.240	0.242	0.816	-0.001	-0.002	0.026
75	3.501	0.198	0.197	0.215	0.012	0.041	0.025
90	3.499	0.180	0.180	0.265	0.035	0.044	0.022
95	3.501	0.175	0.175	0.313	0.056	-0.015	0.024
98	3.500	0.173	0.173	0.458	0.017	0.018	0.020
99	3.500	0.172	0.172	0.336	-0.009	0.006	0.020
100	3.503	0.172	0.171	0.699	0.014	-0.057	0.022
101	3.500	0.172	0.170	1.224	0.007	-0.064	0.022
102	3.499	0.168	0.169	0.930	-0.011	0.048	0.020
105	3.499	0.168	0.167	0.918	0.022	-0.039	0.020
110	3.499	0.162	0.163	0.570	-0.009	-0.024	0.020
150	3.501	0.141	0.139	1.174	0.014	-0.158	0.025
200	3.501	0.122	0.121	0.959	0.001	-0.042	0.016
300	3.500	0.098	0.099	0.488	0.008	0.081	0.017
500	3.500	0.077	0.076	0.894	-0.030	0.016	0.015

Theoretical expected mean: 3.5

At the selected n , empirical mean = 3.49467; expected mean = 3.50000. Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
9	0.013	0.027	-0.197	-0.160	0.062	0.060
10	0.000	0.048	-0.146	-0.127	0.057	0.058
12	0.010	-0.013	-0.154	-0.192	0.052	0.054

03 / Exponential

Required investigation. Selected n: 90. Practical judgment; borderline neighbors.

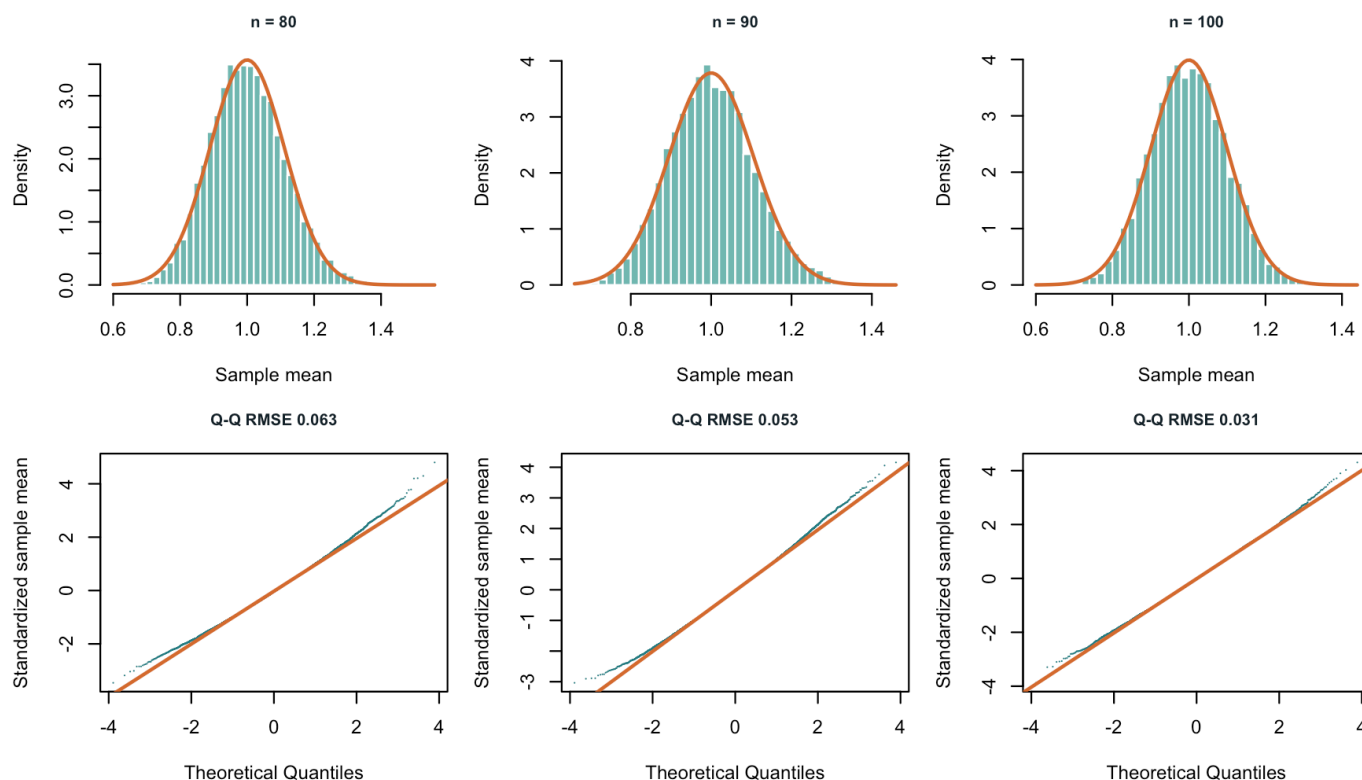
Population, assumptions and theory

Independent Exponential(rate=1) observations: nonnegative with a long right tail. Mean 1, SD 1; $SE=1/\sqrt{n}$. The average has Gamma shape n and rate n, skewness $2/\sqrt{n}$, and excess kurtosis $6/n$.

Retrospective theory-based expectation: Right skew should diminish gradually; $n=30$ need not give accurate tails even when center and SE already agree with theory.

Boundary evidence

Fresh samples, primary seed 7616, B = 10,000 at each n. Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: Bell-shaped central mass with modest right skew; both seeds give Q-Q RMSE about 0.053-0.054 and SE agrees with theory.

Why I did not choose smaller tested n: At $n=30-60$ right-tail bending is clearer; 75-80 are borderline and less consistent across seeds. 90 is a compromise, not a sharp threshold.

Issue / noteworthy finding: Good central approximation does not guarantee accurate extreme tails.

Exponential / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	1.005	1.002	1.000	0.159	1.992	5.936	0.438
2	1.001	0.717	0.707	1.338	1.478	3.665	0.328
3	1.001	0.574	0.577	0.624	1.089	1.638	0.257
5	0.994	0.443	0.447	1.038	0.886	1.117	0.208
10	1.000	0.316	0.316	0.227	0.600	0.582	0.140
15	0.995	0.253	0.258	1.841	0.465	0.237	0.113
20	1.001	0.222	0.224	0.567	0.451	0.322	0.106
30	1.002	0.182	0.183	0.297	0.318	0.128	0.076
50	0.998	0.142	0.141	0.462	0.285	0.060	0.069
60	1.002	0.131	0.129	1.154	0.249	0.119	0.060
75	1.000	0.115	0.115	0.778	0.248	-0.028	0.062
80	1.002	0.113	0.112	0.721	0.265	0.125	0.063
90	1.002	0.106	0.105	0.899	0.214	0.032	0.053
100	1.000	0.100	0.100	0.035	0.119	0.029	0.031
110	1.002	0.096	0.095	0.514	0.219	0.036	0.054
150	1.001	0.082	0.082	0.045	0.159	0.036	0.039
200	1.000	0.070	0.071	0.424	0.124	-0.050	0.033
290	1.000	0.059	0.059	0.389	0.115	0.030	0.029
295	1.001	0.058	0.058	0.066	0.100	0.008	0.026
298	1.000	0.058	0.058	0.176	0.101	0.133	0.028
299	1.001	0.058	0.058	0.633	0.164	0.129	0.041
300	1.001	0.058	0.058	0.430	0.169	0.026	0.041
301	0.999	0.058	0.058	0.933	0.140	-0.056	0.037
302	1.000	0.056	0.058	1.897	0.116	0.069	0.030
305	1.001	0.058	0.057	0.626	0.116	0.043	0.031
310	1.001	0.057	0.057	0.273	0.137	0.051	0.033
500	1.000	0.045	0.045	0.397	0.076	0.002	0.021
1000	1.000	0.032	0.032	0.296	0.020	0.035	0.011

Theoretical expected mean: 1

At the selected n , empirical mean = 1.00170; expected mean = 1.00000. Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
80	0.265	0.205	0.125	0.089	0.063	0.049
90	0.214	0.224	0.032	0.091	0.053	0.054
100	0.119	0.172	0.029	-0.004	0.031	0.043

04 / Binomial small

Required investigation. Selected n: 75. Practical judgment; borderline neighbors.

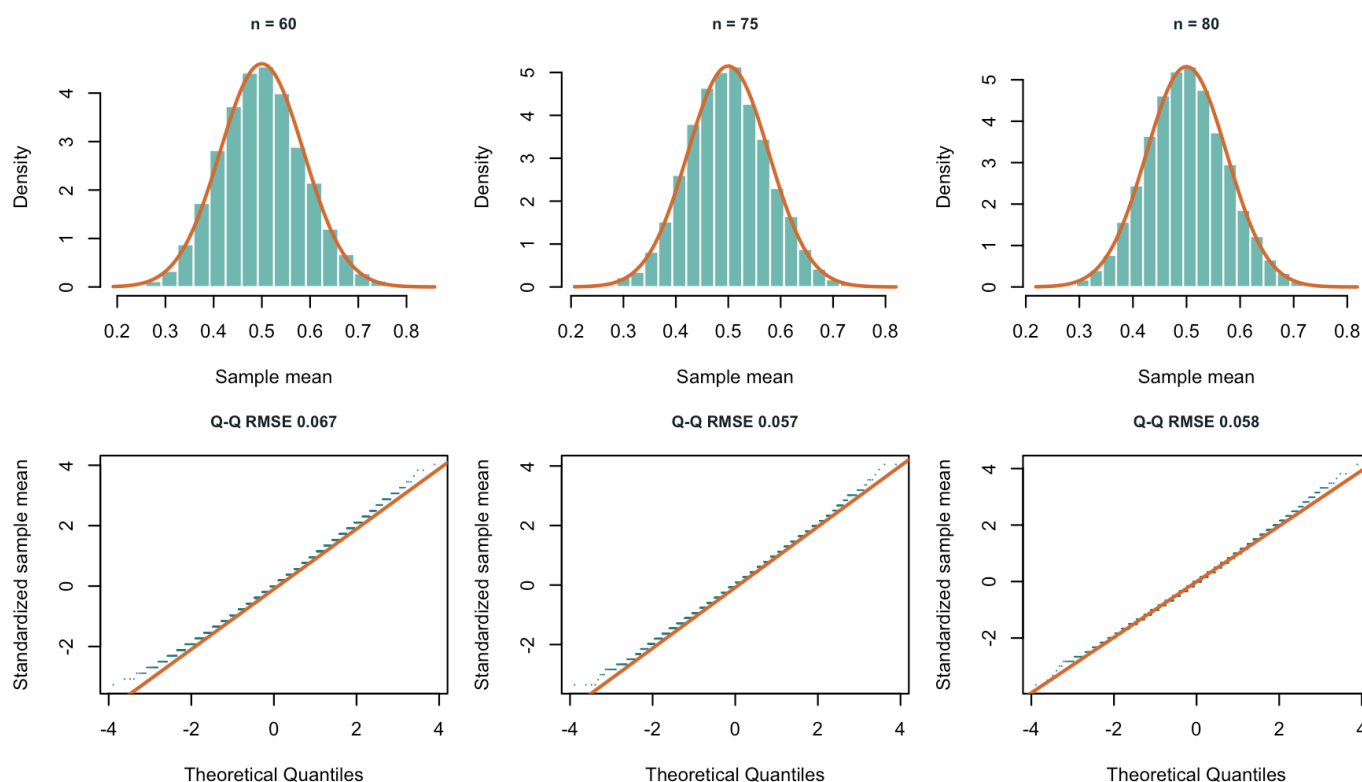
Population, assumptions and theory

Independent $Y \sim \text{Binomial}(\text{size}=5, p=.10)$ observations: discrete, mostly small counts, right-skewed. Mean .5, variance .45, SD $\sqrt{.45}$. Binomial size=5 is fixed; n counts how many such Y values are averaged. Their sum is $\text{Binomial}(5n, .10)$.

Retrospective theory-based expectation: Small p produces skew and lattice steps. Increasing n increases the number of underlying Bernoulli trials and makes the Normal approximation more useful.

Boundary evidence

Fresh samples, primary seed 7616, B = 10,000 at each n. Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: Modest skew and broadly linear Q-Q plot apart from steps; Q-Q RMSE about 0.057-0.060 across seeds and correct SE.

Why I did not choose smaller tested n: At $n=20-50$ skew and steps are clearer; 60 remains borderline with Q-Q RMSE about 0.067 in both seeds. I do not require steps to disappear.

Issue / noteworthy finding: Binomial size=5 is a population parameter, not sample size n.

Binomial small / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	0.501	0.674	0.671	0.470	1.189	0.946	0.563
2	0.494	0.467	0.474	1.502	0.827	0.433	0.408
3	0.496	0.388	0.387	0.133	0.706	0.397	0.328
5	0.504	0.303	0.300	0.955	0.569	0.350	0.245
10	0.499	0.211	0.212	0.747	0.359	-0.005	0.166
15	0.500	0.171	0.173	1.460	0.306	0.045	0.136
20	0.500	0.150	0.150	0.055	0.295	0.169	0.120
30	0.501	0.123	0.122	0.122	0.248	-0.028	0.101
50	0.499	0.095	0.095	0.268	0.163	-0.035	0.074
60	0.500	0.087	0.087	0.234	0.153	-0.045	0.067
75	0.500	0.077	0.077	0.041	0.109	0.050	0.057
80	0.500	0.075	0.075	0.384	0.138	0.046	0.058
90	0.501	0.071	0.071	1.090	0.133	0.141	0.057
100	0.500	0.067	0.067	0.040	0.098	-0.025	0.050
110	0.501	0.065	0.064	1.392	0.111	0.058	0.049
150	0.500	0.054	0.055	0.540	0.088	-0.023	0.042
200	0.500	0.048	0.047	0.552	0.077	0.005	0.038
290	0.500	0.040	0.039	0.481	0.071	-0.029	0.032
295	0.500	0.039	0.039	0.767	0.072	-0.039	0.033
298	0.500	0.039	0.039	1.095	0.034	0.019	0.028
299	0.501	0.038	0.039	1.175	0.099	-0.020	0.036
300	0.501	0.039	0.039	1.168	0.100	-0.027	0.037
301	0.500	0.039	0.039	0.835	0.095	-0.012	0.035
302	0.499	0.038	0.039	0.817	0.114	0.040	0.038
305	0.500	0.038	0.038	0.303	0.030	0.006	0.029
310	0.500	0.039	0.038	1.251	0.041	0.005	0.028
500	0.500	0.030	0.030	1.034	0.071	0.001	0.028
750	0.500	0.025	0.024	0.032	0.066	0.001	0.025
1000	0.500	0.021	0.021	0.292	0.060	0.035	0.022

Theoretical expected mean: 0.5

At the selected n , empirical mean = 0.49960; expected mean = 0.50000. Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
60	0.153	0.149	-0.045	0.003	0.067	0.067
75	0.109	0.136	0.050	0.058	0.057	0.060
80	0.138	0.129	0.046	-0.040	0.058	0.059

05 / Cauchy

Required investigation. Selected n: no finite n, by theory.

Population, assumptions and theory

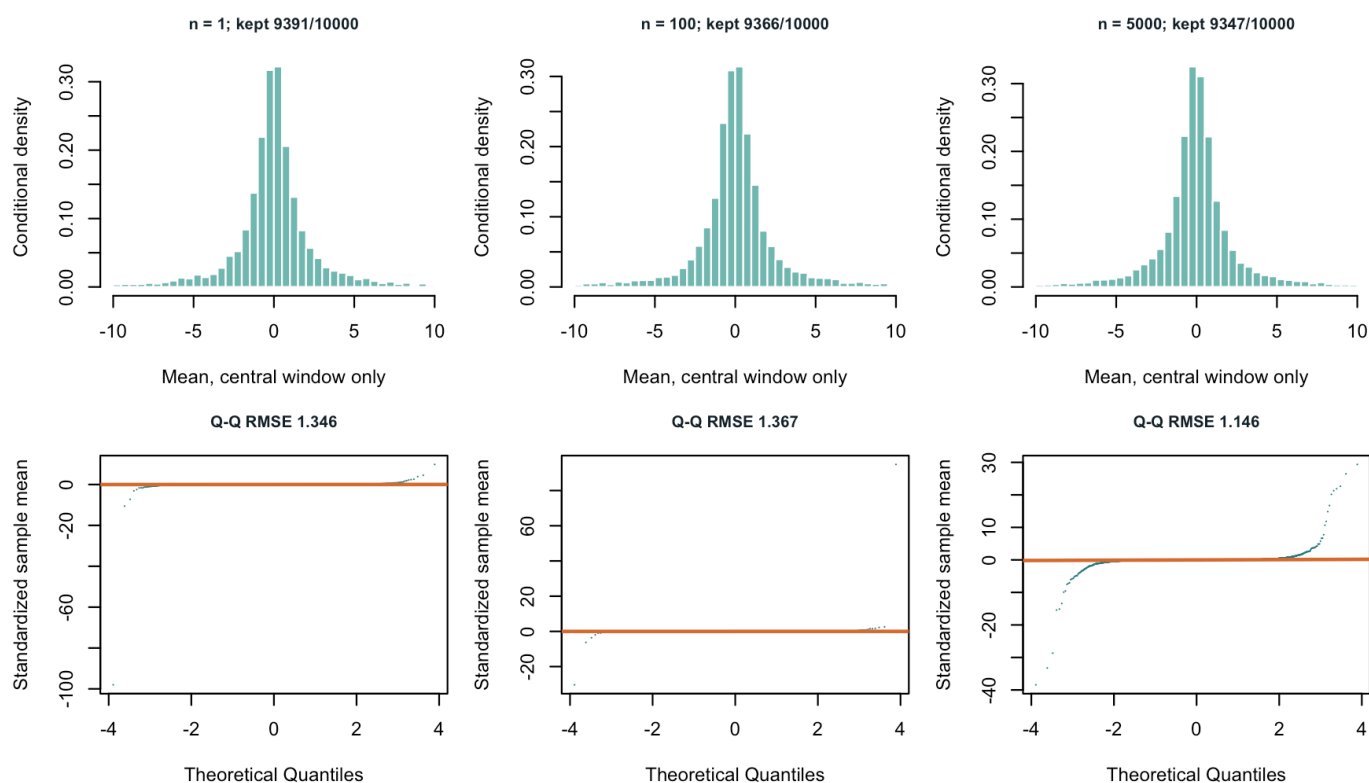
Independent standard Cauchy observations (location 0, scale 1): symmetric with exceptionally heavy tails. Location/median 0 is NOT a population mean. Population mean, variance and SD do not exist. Neither $\sigma/\sqrt{n}$ nor an ordinary finite-variance CLT applies.

Retrospective theory-based expectation: Increasing n should not create a Normal mean distribution. An occasional stable-looking running mean would not establish convergence.

Boundary evidence

Fresh samples, primary seed 7616, $B = 10,000$ at each n . Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.

Cauchy histograms display only means between -10 and 10 , with density normalized within that window; the number retained is labeled. Q-Q plots and all diagnostics use ALL 10,000 means, including extreme tails.



I chose this n because: No finite selection: iid standard Cauchy averages remain standard Cauchy.

Why I did not choose smaller tested n: Increasing n cannot repair the missing finite mean and variance; the ordinary CLT does not apply.

Issue / noteworthy finding: Not merely slow: no Normal convergence. Tested through $n=5000$.

Cauchy / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	-1.920	189.509	N/A	N/A	-94.295	9242.826	1.346
2	0.808	64.402	N/A	N/A	53.429	4164.741	1.267
3	-2.830	257.534	N/A	N/A	-65.623	5276.801	1.328
5	0.060	83.244	N/A	N/A	-5.531	2195.864	1.264
10	2.491	189.412	N/A	N/A	38.813	2816.820	1.315
15	2.123	90.113	N/A	N/A	64.065	5177.672	1.279
20	-2.848	262.954	N/A	N/A	-96.844	9579.369	1.354
30	0.757	119.903	N/A	N/A	8.931	3361.578	1.305
50	1.088	107.703	N/A	N/A	32.081	2173.747	1.280
75	1.530	100.092	N/A	N/A	61.138	4818.269	1.281
100	7.844	1191.395	N/A	N/A	82.471	8173.811	1.367
150	7.643	737.606	N/A	N/A	98.967	9860.626	1.374
200	2.687	401.061	N/A	N/A	95.373	9423.265	1.364
300	13.271	1433.873	N/A	N/A	97.859	9731.531	1.377
490	-0.272	69.911	N/A	N/A	-31.886	3317.398	1.260
495	1.948	158.046	N/A	N/A	47.280	3164.958	1.303
498	-0.163	63.951	N/A	N/A	-28.803	2495.664	1.256
499	0.912	112.885	N/A	N/A	73.920	6780.786	1.308
500	-2.127	101.349	N/A	N/A	-42.514	2682.600	1.279
501	0.597	77.038	N/A	N/A	6.648	1954.940	1.251
502	-1.823	97.080	N/A	N/A	-62.196	4657.091	1.289
505	1.801	191.487	N/A	N/A	80.960	7722.006	1.326
510	1.951	239.637	N/A	N/A	94.478	9274.066	1.346
1000	11.338	1052.443	N/A	N/A	99.759	9967.551	1.378
2000	-0.139	75.699	N/A	N/A	-22.845	3286.919	1.268
5000	0.060	32.534	N/A	N/A	-3.468	649.912	1.146

Theoretical expected mean: N/A: undefined for Cauchy; not supplied analytically for the dependent process.

The large finite empirical means, SEs and moment diagnostics are descriptive only: they do not estimate existing Cauchy population moments. All tested averages have the same Cauchy law by theory.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
1	-94.295	80.887	9242.826	6904.558	1.346	1.361
100	82.471	-34.431	8173.811	1774.658	1.367	1.273
5000	-3.468	-11.339	649.912	3227.944	1.146	1.276

06 / Dependent failure

Extra-credit investigation. Selected n: 350. Empirical judgment; dependent CLT not proved.

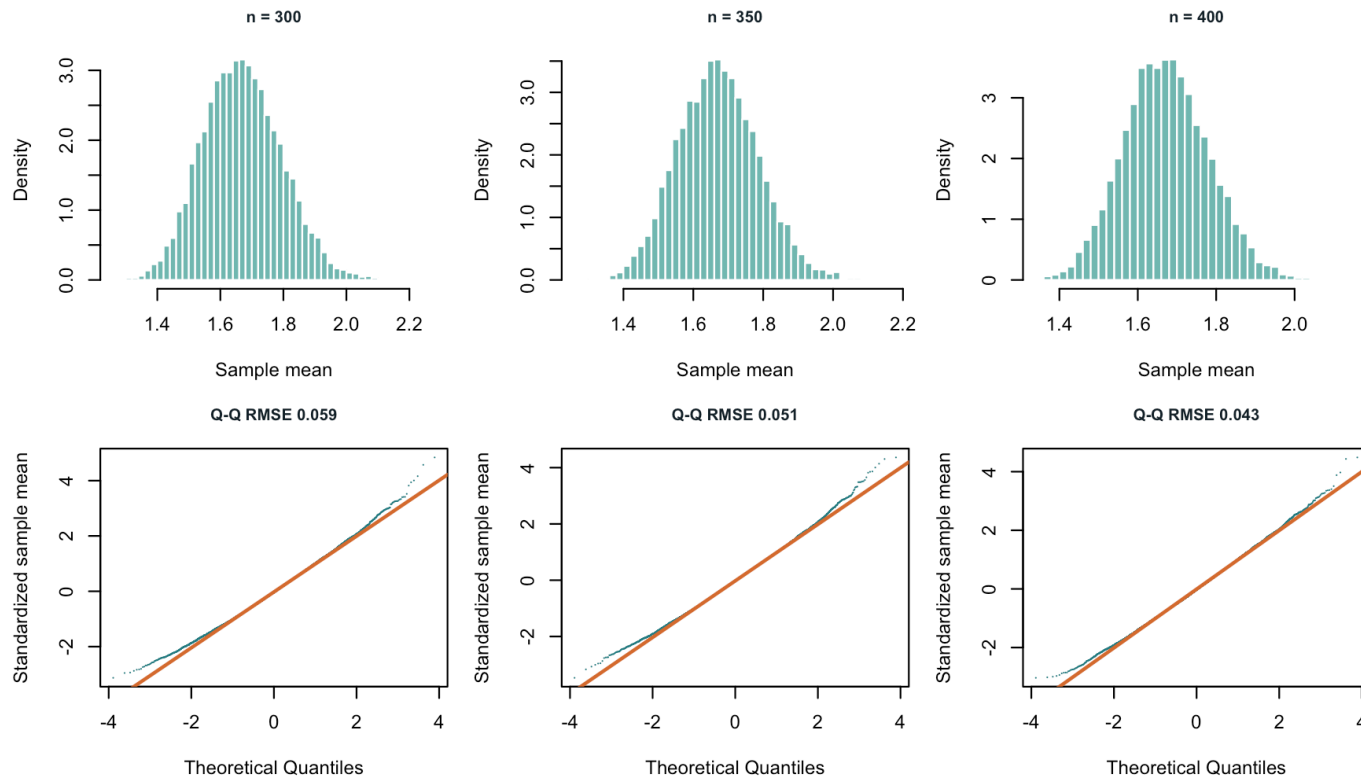
Population, assumptions and theory

Instructor recurrence: first operating time is $\text{Exponential}(\text{rate}=1)$; later $\text{rate}=\exp(1.25 \cdot (\text{previous time}-1))$, followed by an Exponential draw at that rate. Each Monte Carlo sample restarts this process; no burn-in is silently added. Times inside a sample are dependent and their transient marginals are not identical. Baseline mean 1 is not a theoretical mean for the full process.

Retrospective theory-based expectation: The compensating rate recurrence should change the mean distribution and covariance. Approximate Normality is plausible empirically, but the iid SE formula is not justified.

Boundary evidence

Fresh samples, primary seed 7616, $B = 10,000$ at each n . Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: Bell-shaped empirical mean distribution with modest right skew; Q-Q RMSE about 0.049-0.051 across seeds and comparable larger neighbors.

Why I did not choose smaller tested n: At $n=100-200$ curvature is clearer; 300 is borderline with more skew. 350 is an empirical compromise, not a proved dependent CLT.

Issue / noteworthy finding: No iid theoretical SE; covariance matters. Exact boundary and asymptotic behavior remain uncertain.

Dependent failure / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	1.015	1.010	N/A	N/A	1.919	5.344	0.433
2	1.272	0.965	N/A	N/A	2.625	11.039	0.482
3	1.348	0.873	N/A	N/A	2.187	7.388	0.442
5	1.484	0.809	N/A	N/A	1.824	4.918	0.392
10	1.576	0.624	N/A	N/A	1.254	2.134	0.293
15	1.608	0.527	N/A	N/A	1.068	1.711	0.248
20	1.631	0.465	N/A	N/A	0.883	1.044	0.209
30	1.641	0.384	N/A	N/A	0.683	0.568	0.164
50	1.655	0.306	N/A	N/A	0.592	0.484	0.140
75	1.658	0.247	N/A	N/A	0.483	0.320	0.114
100	1.669	0.213	N/A	N/A	0.408	0.232	0.097
150	1.669	0.174	N/A	N/A	0.334	0.176	0.079
200	1.669	0.154	N/A	N/A	0.310	0.162	0.074
300	1.670	0.125	N/A	N/A	0.241	0.052	0.059
350	1.672	0.115	N/A	N/A	0.208	0.097	0.051
400	1.673	0.108	N/A	N/A	0.170	-0.003	0.043
450	1.673	0.102	N/A	N/A	0.170	-0.001	0.043
500	1.673	0.098	N/A	N/A	0.161	-0.027	0.041
550	1.673	0.092	N/A	N/A	0.143	-0.001	0.037
740	1.674	0.080	N/A	N/A	0.159	0.011	0.040
745	1.674	0.080	N/A	N/A	0.127	0.011	0.031
748	1.674	0.079	N/A	N/A	0.227	0.145	0.055
749	1.675	0.078	N/A	N/A	0.112	0.035	0.028
750	1.672	0.080	N/A	N/A	0.175	0.050	0.042
751	1.673	0.080	N/A	N/A	0.173	0.072	0.043
752	1.673	0.080	N/A	N/A	0.169	0.064	0.041
755	1.672	0.079	N/A	N/A	0.137	0.014	0.034
760	1.673	0.078	N/A	N/A	0.109	0.010	0.027
1000	1.674	0.069	N/A	N/A	0.147	0.095	0.036

Theoretical expected mean: N/A: undefined for Cauchy; not supplied analytically for the dependent process.

At the selected n , empirical mean = 1.67162; expected mean = N/A. Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
300	0.241	0.233	0.052	-0.025	0.059	0.058
350	0.208	0.197	0.097	-0.005	0.051	0.049
400	0.170	0.174	-0.003	-0.018	0.043	0.044

07 / Basketball shots

Extra-credit investigation. Selected n: 140. Practical judgment; borderline neighbors.

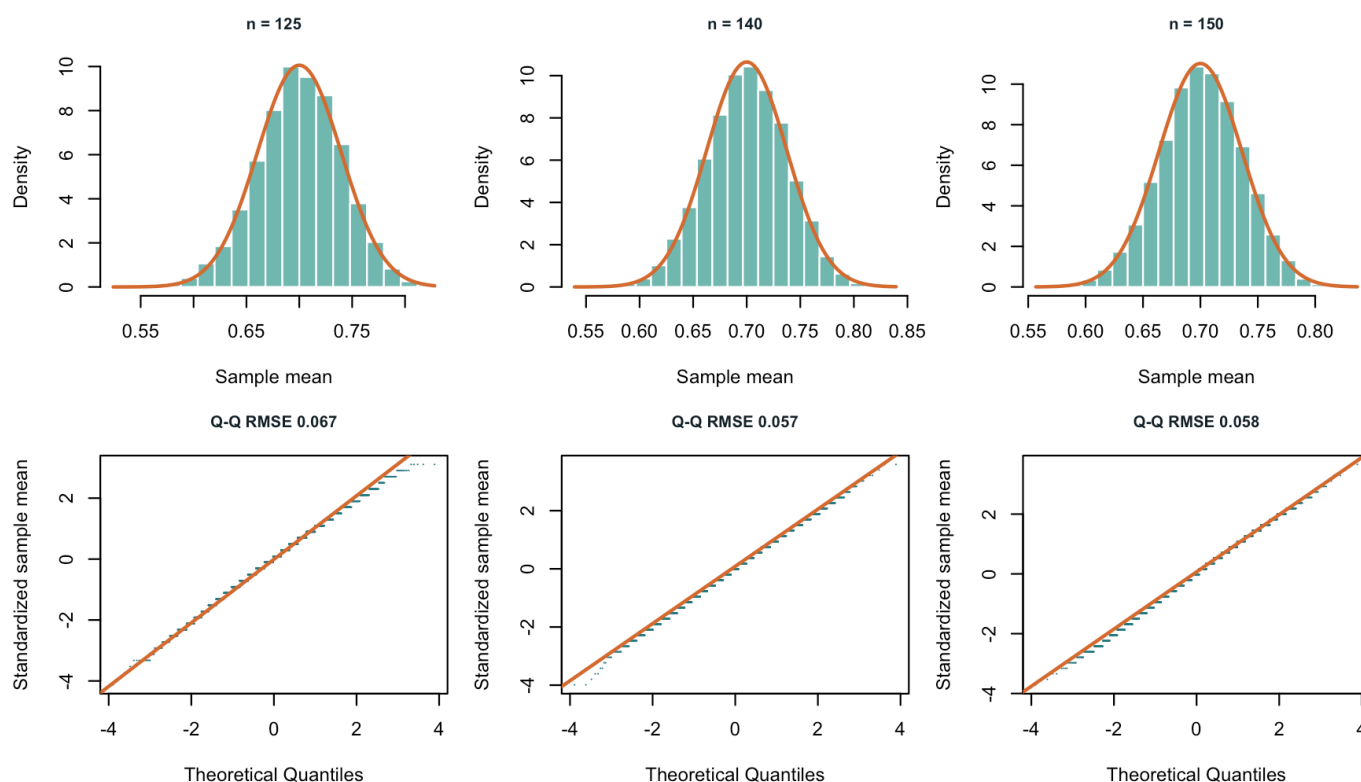
Population, assumptions and theory

Independent Bernoulli shots, with n probabilities equally spaced from .90 to .50. For $n=1$ the probability is .90; for $n \geq 2$ the expected average is .70. The total is Poisson-binomial, not generally Binomial($n, .70$). Changing n constructs a new probability schedule rather than taking a prefix of one fixed schedule.

Retrospective theory-based expectation: Independent bounded summands can give a Normal approximation without identical distributions, with visible lattice steps at modest n and the correct varying- p SE.

Boundary evidence

Fresh samples, primary seed 7616, $B = 10,000$ at each n . Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: Small skew; the main departure is fine discrete steps. Q-Q RMSE about 0.057-0.061 across seeds; SE matches the varying-probability formula.

Why I did not choose smaller tested n : At $n=30-100$ gaps are clearer; 110-125 are borderline with Q-Q RMSE about 0.064-0.067. 140 is my first convincing compromise.

Issue / noteworthy finding: Independent is not identically distributed; sum the individual Bernoulli variances.

Basketball shots / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	0.898	0.303	0.300	0.844	-2.632	4.930	0.908
2	0.702	0.294	0.292	0.723	-0.396	-0.705	0.536
3	0.701	0.248	0.247	0.178	-0.393	-0.444	0.430
5	0.701	0.194	0.195	0.492	-0.332	-0.228	0.325
10	0.700	0.139	0.139	0.138	-0.226	-0.155	0.220
15	0.701	0.114	0.114	0.284	-0.205	-0.018	0.178
20	0.700	0.099	0.099	0.263	-0.177	-0.120	0.154
30	0.700	0.081	0.081	0.250	-0.139	-0.066	0.125
50	0.699	0.062	0.063	1.284	-0.076	-0.071	0.096
75	0.699	0.052	0.051	1.393	-0.073	0.013	0.077
100	0.700	0.045	0.044	0.489	-0.119	-0.030	0.072
110	0.700	0.042	0.042	0.166	-0.088	-0.018	0.067
125	0.700	0.040	0.040	0.413	-0.135	0.012	0.067
140	0.700	0.038	0.037	0.619	-0.052	-0.039	0.057
150	0.701	0.036	0.036	0.064	-0.072	-0.086	0.058
160	0.700	0.035	0.035	0.509	-0.039	0.030	0.054
200	0.700	0.031	0.031	1.289	-0.059	-0.027	0.050
300	0.700	0.025	0.026	0.566	-0.045	-0.026	0.041
490	0.700	0.020	0.020	0.439	-0.035	0.062	0.034
495	0.700	0.020	0.020	0.349	-0.058	-0.022	0.034
498	0.700	0.020	0.020	0.744	0.005	-0.043	0.032
499	0.700	0.020	0.020	0.873	-0.035	-0.019	0.034
500	0.700	0.020	0.020	1.216	-0.016	0.021	0.032
501	0.700	0.020	0.020	0.266	-0.044	-0.040	0.033
502	0.700	0.020	0.020	1.298	-0.047	0.053	0.033
505	0.700	0.020	0.020	0.474	-0.084	-0.048	0.038
510	0.700	0.020	0.020	0.831	-0.076	0.067	0.035
750	0.700	0.016	0.016	0.162	-0.022	-0.080	0.028
1000	0.700	0.014	0.014	0.325	-0.005	0.014	0.023

Theoretical expected mean: .90 for $n=1$; .70 for every tested $n \geq 2$.

At the selected n , empirical mean = 0.70035; expected mean = 0.70000. Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
125	-0.135	-0.110	0.012	0.002	0.067	0.064
140	-0.052	-0.099	-0.039	0.075	0.057	0.061
150	-0.072	-0.062	-0.086	-0.048	0.058	0.056

08 / Synthetic ages

Extra-credit investigation. Selected n: 15. Practical judgment; borderline neighbors.

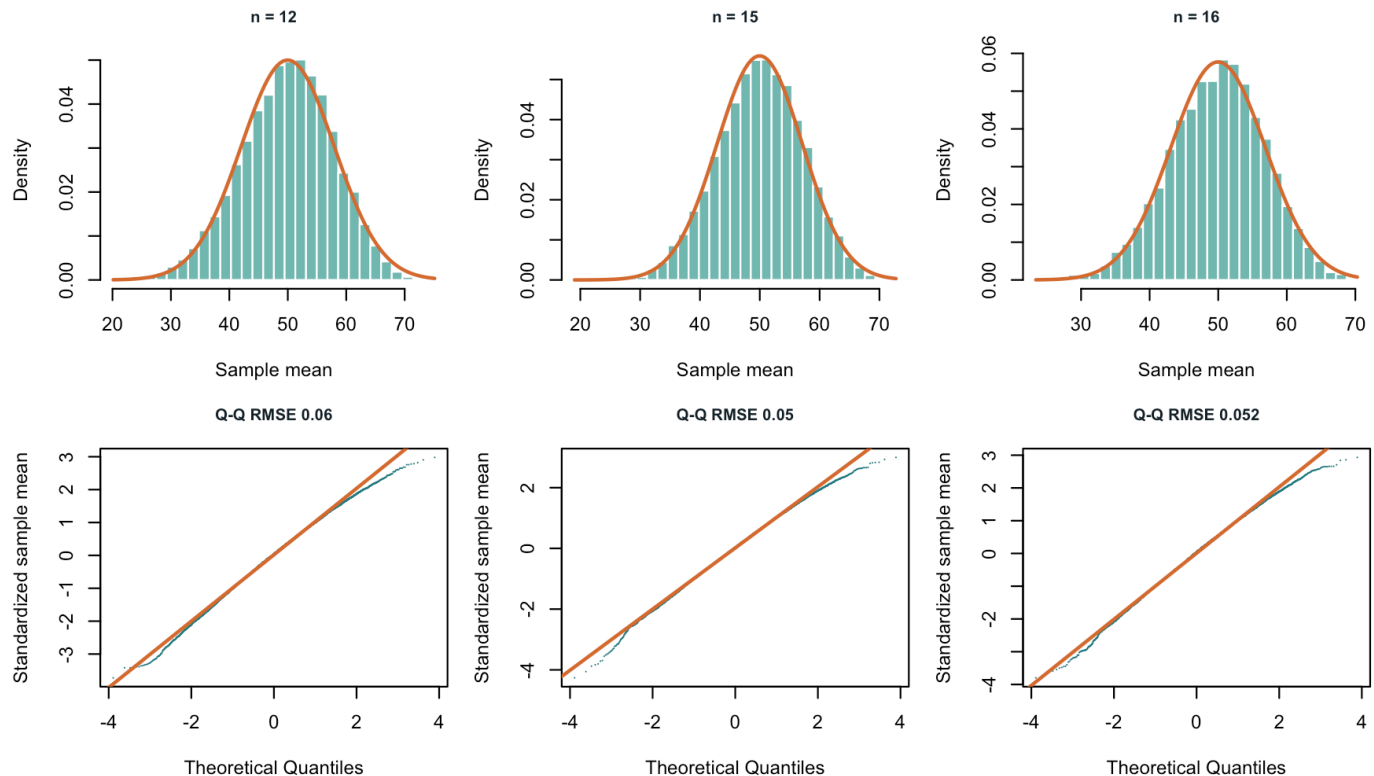
Population, assumptions and theory

Instructor synthetic mixture on ages 0:100: weights $.045\exp(-age/5) + .030\exp(-.5*((age-65)/10)^2)$, normalized to sum one. There is an early-age spike and a broad adult peak. Mean and SD are computed exactly from the discrete probabilities: 49.981 and 27.635. This is NOT historical mortality data.

Retrospective theory-based expectation: Finite variance and bounded ages support the iid CLT, but the two-component shape may be conspicuous for small n. A complex population shape need not imply the slowest convergence.

Boundary evidence

Fresh samples, primary seed 7616, B = 10,000 at each n. Top: sample-mean histograms; bottom: standardized Normal Q-Q plots. Orange histogram curves are theoretical Normal approximations only where valid; orange Q-Q lines are fitted quartile references.



I chose this n because: The irregular mixture has smoothed to a bell shape with modest left skew; Q-Q RMSE about 0.050-0.056 across seeds and correct SE.

Why I did not choose smaller tested n: At n=5-10 asymmetry is clearer; 12 is borderline with a stronger left-tail bend. 15 is acceptable, although larger n look better.

Issue / noteworthy finding: Synthetic, not historical mortality data; a complicated shape need not converge slowly.

Synthetic ages / numerical evidence

Search design: broad grid first, then population-specific refinements and smaller-boundary checks. The complete tested grid below makes the choice auditable; retained large-n points check longer-range behavior, not a requirement that every case use the same refinement.

All rows use $B = 10,000$, seed 7616. SE is the SD of simulated means. Kurt. is excess kurtosis. Relative SE error is reported as a percentage; N/A is not zero.

n	Mean	SE	SE theory	Error %	Skew	Kurt.	QQ
1	50.065	27.612	27.635	0.080	-0.872	-0.770	0.441
2	49.970	19.492	19.541	0.246	-0.599	-0.411	0.268
3	49.942	15.885	15.955	0.434	-0.489	-0.286	0.187
5	50.056	12.357	12.359	0.013	-0.371	-0.209	0.116
10	49.955	8.815	8.739	0.873	-0.279	-0.011	0.071
12	49.898	7.895	7.977	1.031	-0.237	-0.059	0.060
15	50.052	7.128	7.135	0.100	-0.191	-0.039	0.050
16	49.860	6.891	6.909	0.258	-0.202	-0.062	0.052
18	49.903	6.523	6.514	0.146	-0.136	-0.099	0.038
20	49.917	6.174	6.179	0.087	-0.182	0.043	0.044
22	49.934	5.859	5.892	0.559	-0.168	-0.097	0.045
30	49.962	5.029	5.045	0.317	-0.110	-0.064	0.030
50	49.994	3.950	3.908	1.060	-0.123	-0.077	0.033
75	49.951	3.184	3.191	0.227	-0.085	-0.079	0.025
90	49.967	2.898	2.913	0.499	-0.121	0.067	0.032
95	50.011	2.861	2.835	0.907	-0.088	0.020	0.024
98	49.977	2.811	2.792	0.681	-0.063	-0.110	0.023
99	49.937	2.786	2.777	0.325	-0.105	0.091	0.030
100	49.968	2.770	2.763	0.244	-0.074	-0.009	0.020
101	50.015	2.723	2.750	0.985	-0.114	0.071	0.029
102	49.950	2.773	2.736	1.333	-0.026	-0.099	0.018
105	49.980	2.697	2.697	0.001	-0.071	-0.032	0.021
110	49.964	2.656	2.635	0.811	-0.100	-0.040	0.027
150	49.963	2.252	2.256	0.207	-0.083	0.053	0.024
200	49.986	1.948	1.954	0.290	-0.076	0.031	0.022
300	49.998	1.574	1.595	1.355	-0.099	0.058	0.028
500	49.991	1.236	1.236	0.015	-0.038	0.015	0.014

Theoretical expected mean: 49.98098

At the selected n, empirical mean = 50.05187; expected mean = 49.98098. Where valid theoretical references exist, center/SE agreement supports implementation but cannot by itself establish Normal shape.

Fixed-seed sensitivity at the boundary:

n	Skew 7616	Skew 7617	Kurt. 7616	Kurt. 7617	QQ 7616	QQ 7617
12	-0.237	-0.215	-0.059	-0.103	0.060	0.058
15	-0.191	-0.201	-0.039	-0.159	0.050	0.056
16	-0.202	-0.197	-0.062	-0.058	0.052	0.050

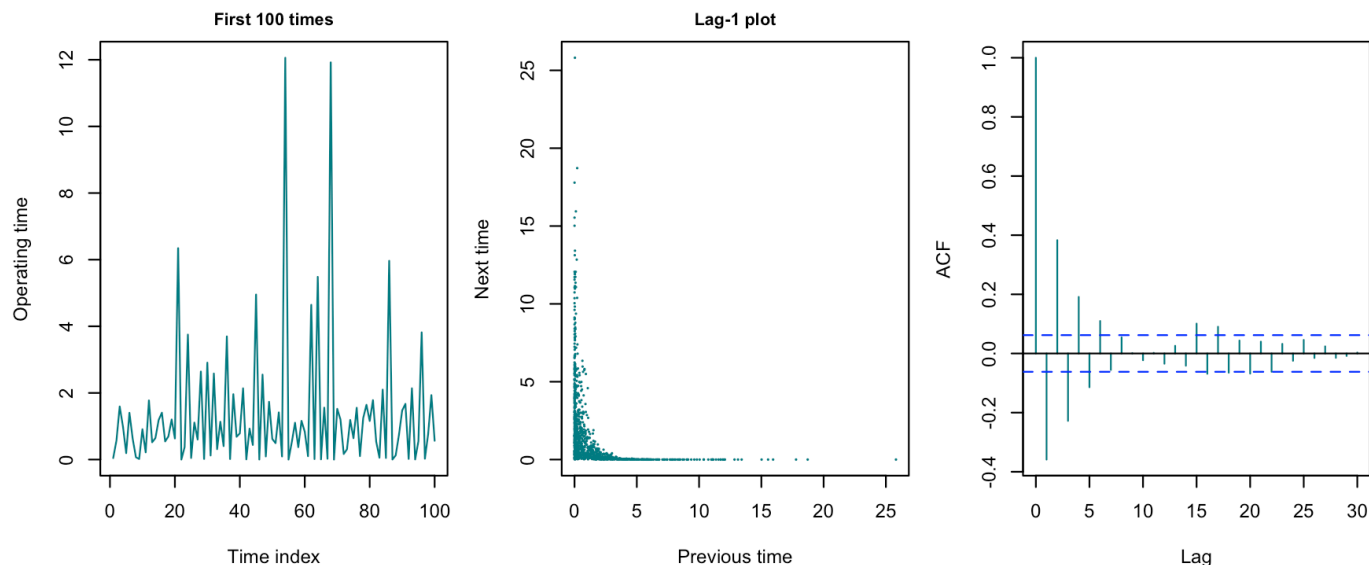
Special-process checks

Cauchy: theory distinguishes impossible from merely slow

The standard Cauchy characteristic function is $\exp(-|t|)$. For independent observations, the mean has characteristic function $[\exp(-|t|/n)]^n = \exp(-|t|)$, the same Cauchy distribution for every n . Thus even arbitrarily large n cannot yield a Normal mean distribution. Symmetry alone is not enough.

The running mean on page 3 shows one path, not the sampling distribution. Occasional extreme observations can dominate it. Empirical SD-standardized Q-Q RMSE is numerically defined for each finite run, but Cauchy has no population SD and this metric is especially unstable. It is not a valid theoretical z-score or a convergence proof.

Dependent failure: verify the recurrence, not iid draws



This particular 1,000-time path has lag-1 correlation -0.359 (seed 7616); that is a process diagnostic, not an analytic covariance model. Larger previous times raise the next rate and shorten its conditional expected time. Since operating times are nonnegative, rate is at least $\exp(-1.25)$, and conditional second moments are at most $2 \exp(2.5)$. Finite individual moments do not establish the mixing/stationarity conditions needed for a dependent CLT. The empirical choice 350 remains qualified; its asymptotic justification is unresolved, unlike Cauchy's proved non-convergence.

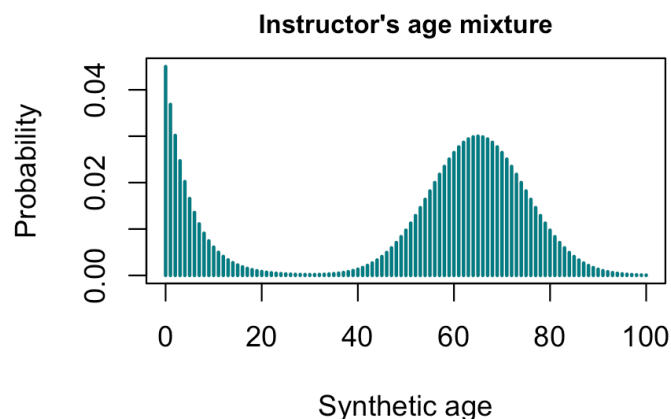
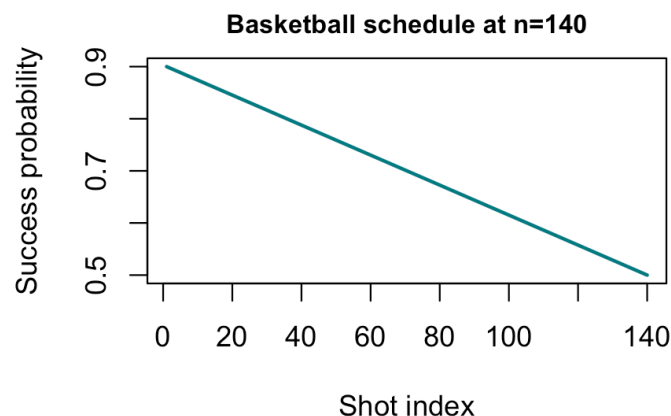
Extra-credit construction checks

Basketball: correct mean and variance, not an iid shortcut

$$E(\bar{X}) = \frac{1}{n} \sum_i p_i, \quad \text{Var}(\bar{X}) = \frac{1}{n^2} \sum_i p_i(1 - p_i).$$

For $n \geq 2$ the equally spaced probabilities average .70. Replacing every p_i by $\bar{p}$ ignores heterogeneity: $\bar{p}(1-\bar{p})/n$ exceeds the correct variance by $\text{mean}((p_i - \bar{p})^2)/n$. Independence makes covariances zero; it does not make probabilities equal. Bounded summands and a growing total variance support a generalized independent-summand CLT.

n	Expected mean	Correct SE	iid shortcut SE
1	0.9	0.3000	0.3000
30	0.7	0.0808	0.0837
125	0.7	0.0396	0.0410
140	0.7	0.0375	0.0387
150	0.7	0.0362	0.0374



Age mixture: preserve the supplied synthetic model

I normalize the two instructor weight components over integer ages 0–100, then sample with replacement using those probabilities. Exact finite-population moments come from $\text{sum}(\text{age} \times \text{probability})$ and $\text{sum}((\text{age} - \text{mean})^2 \times \text{probability})$. No real mortality data, empirical fit or historical claim is introduced. The contrast between this irregular population and its smoother mean distribution at $n=15$ is a noteworthy result, not proof that all mixtures converge quickly.

AI iteration: what changed and why

1 / A complicated implementation needed correction

Actual initial request, summarized: build a professional R + Quarto + GitHub Monte Carlo project with reusable functions, $B=10,000$, seed 7616, all eight cases, multiple diagnostics, Normal calibration, and no invented conclusions.

Useful AI response: a reusable simulator, generators, diagnostic functions and separate reports. **Weakness identified during AI audit:** the alternate-seed experiment could overwrite saved primary means, so a plot might not represent its CSV row. Shared refinement values also created unnecessary experiments across unrelated populations. These were AI-discovered technical defects; I do not claim I personally found the code bugs.

Actual follow-up, summarized: after sharing the professor's rubric, I asked for a plan emphasizing a clear summary and defensible reasons, then requested implementation of the plan including primary/sensitivity separation and population-specific refinement.

Improvement: separate seed directories; each plotted vector is checked against its corresponding row for B , mean, SD and Q-Q RMSE. Targeted smaller- n experiments replace relying on oversized choices. **Check:** the R validation script checks both seeds and the model constructions. A fresh-directory run reproduced saved result CSVs byte-for-byte on the same R version using the dependency cache. This is a computational check, not an independent student's statistical review.

2 / "Looks better" did not answer the smallest- n question

Actual discussion: I initially found die $n=20$ convincing and preferred nicer-looking Exponential/Binomial evidence at 100/200. Then I asked: "They looked better. But I didn't know ... we need the smallest defensible n . Maybe then going off visually isn't right?"

Weakness: a prettier plot at a larger n does not explain why smaller n is unacceptable. Requiring every diagnostic to fall within exact-Normal simulation noise was also too strict for the stated practical question.

Improvement: inspect below/at/above candidate boundaries, compare fixed seeds, retain residual skew and discrete steps honestly, and state borderline ranges. For example, the die at 9 and 10 is quite similar; 10 is a practical choice, not proof that 9 fails. Exponential 90 retains visible curvature and theoretical skewness about .211; selecting it is not declaring exact Normality.

Check: both seeds' numerical rows and the actual saved-vector histograms/Q-Q plots appear in this PDF. The result is a stronger argument with explicit limits, not "better" results obtained by changing seeds until the plots look nice.

AI review and my role

3 / The attractive summary was not a complete submission

Actual follow-up: I asked where the PDF directly answered all three major questions and where it showed my AI exchanges and improvements. I then requested: “EVERYTHING should be in the pdf pertaining to the instructions.”

Weakness: the original four-page report left case-level evidence, code explanation and the fuller AI record in the repository. A professor should not have to reconstruct the argument from separate files.

Improvement: this self-contained version explicitly answers Sample Size, Normality and Statistical Assumptions, then includes all eight cases, complete tested-grid diagnostics, seed comparisons, special-process checks, code walkthroughs and reproduction details. Visual polish serves the evidence rather than replacing it.

What I clarified during the actual review

- I first described n versus B in terms of a small sample and a large “resample replication.” The correction is that **n observations make one sample; B counts fresh simulated samples**, not bootstrap resamples.
- My explanation of exact Normality was uncertain. Independent Normal sums and averages are exactly Normal; the simulation validates that fact, it does not discover it.
- I thought Cauchy’s moments might be undefined “sometimes.” They are undefined properties of the population **always**; finite sample means and SDs can still be calculated.
- I asked whether mean and SE can agree while shape differs. Yes: first/second-moment agreement does not imply matching higher moments or tails.

Ownership and limits: I prompted the AI, challenged the smallest- n reasoning, approved a practical tolerance for small tail bends/steps, and requested corrections and validation. AI selected the working boundary integers under that standard; I did not independently confirm every integer. I do not claim complete line-by-line mastery or a personal prediction made before the original runs. The walkthrough below makes the important computation explainable without hiding these limits.

Noteworthy findings, not invented personal reactions: Cauchy averaging does not help; non-identical shots still approach a useful Normal shape with the right SE; the irregular bounded age mixture is fast; the dependent process cannot borrow an iid SE. These are results supported by theory and generated evidence. No unrecorded surprise or personal understanding is attributed to me.

Important R code / one simulator

```
simulate_sample_means <- function (generator, n, B = monte_carlo_reps)
{
  if (length(n) != 1L || n < 1L || n != as.integer(n)) {
    stop("n must be a positive integer", call. = FALSE)
  }
  if (length(B) != 1L || B < 1L || B != as.integer(B)) {
    stop("B must be a positive integer", call. = FALSE)
  }
  vapply(seq_len(B), function(i) mean(generator(as.integer(n))),
    numeric(1))
}
```

`generator(n)` creates one fresh sample of n observations. `mean(...)` returns its one average. `seq_len(B)` supplies B repetition indices; `vapply` runs the same operation B times and requires one numeric result each time. The output is B sample means, not B observations in one sample. The argument checks reject invalid n/B ; the generator carries the population or entire dependent recurrence.

```
set.seed(7616)
# Example of the same reusable simulation call, not new evidence:
means <- simulate_sample_means(exponential_generator, n = 90, B = 10000)
mean(means) # empirical mean of the sampling distribution
sd(means)   # empirical standard error of ONE sample average
```

The production workflow sets seed 7616 immediately before the fixed-order primary experiment, and 7617 before the alternate experiment. Targeted boundary experiments reset their assigned seed before each experiment. The seed and experiment order together define reproducibility; the same seed alone does not make differently ordered experiments identical. No seed shopping is used.

Data separation: primary and alternate vectors are saved in distinct `student_seed` and `student_seed_plus_one` directories. `load_sample_means()` verifies a unique matching CSV row, length B , mean, SD and Q-Q RMSE before drawing a plot. This guards the original overwritten-vector error.

Minimal dependencies: statistical work uses base R. `knitr` and `rmarkdown` support reports; `renv` records/restores package versions. Quarto supplies the Typst PDF renderer. There is no Python statistical analysis and no separate TeX dependency.

Important R code / populations

```
standard_normal_generator <- function (n)
{
  rnorm(n, mean = 0, sd = 1)
}
```

```
six_sided_die_generator <- function (n)
{
  sample(1:6, size = n, replace = TRUE)
}
```

```
exponential_generator <- function (n)
{
  rexp(n, rate = 1)
}
```

```
binomial_small_generator <- function (n)
{
  binomial_size <- 5L
  binomial_probability <- 0.1
  rbinom(n, size = binomial_size, prob = binomial_probability)
}
```

```
cauchy_generator <- function (n)
{
  rcauchy(n, location = 0, scale = 1)
}
```

`rnorm`, `rexp`, `rbinom` and `rcauchy` draw random observations from the specified distributions. The first argument `n` is the number drawn. Die `sample(..., replace=TRUE)` permits repeated independent outcomes. Binomial `size=5` is the number of Bernoulli trials inside each individual observation, not the number of observations averaged. Cauchy location/scale parameters are not a finite mean/SD.

All five generators return a numeric vector of length `n`; the reusable simulator need not know which distribution generated it. Population theoretical moments are supplied separately so undefined Cauchy moments remain N/A.

Important R code / the extra-credit models

```
simulate_dependent_failure <- function (n, baseline_rate = 1, beta = 1.25, reference_time = 1)
{
  if (n < 1L)
    stop("n must be at least 1", call. = FALSE)
  operating_time <- numeric(n)
  failure_rate <- numeric(n)
  failure_rate[1L] <- baseline_rate
  operating_time[1L] <- rexp(1L, rate = failure_rate[1L])
  if (n >= 2L) {
    for (t in 2:n) {
      failure_rate[t] <- baseline_rate * exp(beta * (operating_time[t -
        1L] - reference_time))
      operating_time[t] <- rexp(1L, rate = failure_rate[t])
    }
  }
  operating_time
}
```

```
nonidentical_probabilities <- function (n, probability_start = 0.9, probability_end = 0.5)
{
  seq(probability_start, probability_end, length.out = n)
}
```

```
nonidentical_bernoulli_generator <- function (n)
{
  probabilities <- nonidentical_probabilities(n)
  rbinom(n, size = 1L, prob = probabilities)
}
```

The failure loop uses the **previous generated time**, not a new independent draw, to calculate the next rate. The first time/rate initializes each fresh sample. `if(n>=2)` avoids an invalid loop when `n=1`. For basketball, `seq(..., length.out=n)` builds `n` varying probabilities; vectorized `rbinom(n,size=1,prob=probabilities)` draws independent Bernoulli outcomes with those probabilities.

Important R code / age and shot moments

```
age_at_death_distribution <- function ()
{
  age <- 0:100
  early_component <- 0.045 * exp(-age/5)
  adult_component <- 0.03 * exp(-0.5 * ((age - 65)/10)^2)
  weight <- early_component + adult_component
  probability <- weight/sum(weight)
  data.frame(age = age, probability = probability, percent = 100 *
    probability)
}
```

```
rdeath <- function (n, distribution = age_at_death_distribution())
{
  sample(x = distribution$age, size = n, replace = TRUE, prob = distribution$probability)
}
```

```
age_at_death_theory <- function ()
{
  distribution <- age_at_death_distribution()
  population_mean <- sum(distribution$age * distribution$probability)
  population_variance <- sum((distribution$age - population_mean)^2 *
    distribution$probability)
  list(mean = population_mean, sd = sqrt(population_variance))
}
```

```
nonidentical_bernoulli_theory <- function (n)
{
  probabilities <- nonidentical_probabilities(n)
  list(mean = mean(probabilities), sd = sqrt(sum(probabilities *
    (1 - probabilities))/n^2))
}
```

`weight/sum(weight)` makes a probability distribution. Sampling with `prob=` faithfully reproduces that synthetic population. The weighted mean and squared-deviation sum calculate exact discrete moments. Basketball's returned `sd` is the **SD of the mean** for the specified schedule, not the SD of a single identical Bernoulli population.

Important R code / shape diagnostics

```
sample_skewness <- function (x)
{
  x <- x[is.finite(x)]
  n <- length(x)
  if (n < 3L)
    return(NA_real_)
  centered <- x - mean(x)
  m2 <- mean(centered^2)
  m3 <- mean(centered^3)
  if (m2 == 0)
    return(NA_real_)
  sqrt(n * (n - 1))/(n - 2) * m3/m2^(3/2)
}
```

```
sample_excess_kurtosis <- function (x)
{
  x <- x[is.finite(x)]
  n <- length(x)
  if (n < 4L)
    return(NA_real_)
  centered <- x - mean(x)
  m2 <- mean(centered^2)
  m4 <- mean(centered^4)
  if (m2 == 0)
    return(NA_real_)
  g2 <- m4/m2^2 - 3
  ((n - 1)/((n - 2) * (n - 3))) * ((n + 1) * g2 + 6)
}
```

Centering subtracts the empirical mean. m_2 , m_3 and m_4 are empirical central moments; the factors apply finite-sample adjustments to skewness/excess kurtosis. Here the local n is the vector length **B**, not CLT sample size n . Positive skew means a longer right tail; negative skew a longer left tail. Positive excess kurtosis indicates larger standardized fourth moment than Normal; negative indicates smaller. Neither metric proves Normality.

Important R code / Q-Q and SE error

```
qq_rmse <- function (x)
{
  x <- x[is.finite(x)]
  if (length(x) < 2L || sd(x) == 0)
    return(NA_real_)
  standardized <- sort((x - mean(x))/sd(x))
  theoretical <- qnorm(ppoints(length(standardized)))
  sqrt(mean((standardized - theoretical)^2))
}
```

```
relative_error <- function (estimate, reference)
{
  if (length(reference) != 1L || is.na(reference) || reference ==
    0) {
    return(NA_real_)
  }
  abs(estimate - reference)/abs(reference)
}
```

Q-Q standardization uses empirical SD; `ppoints` avoids quantile endpoints 0/1. Sorting pairs the same ranks in simulated and Normal distributions. Squaring penalizes larger deviations, averaging uses all ranks, and taking a square root returns standardized-distance units. The relative-error denominator is the absolute theoretical reference; undefined/zero references return N/A rather than an invalid percentage.

Finite-value filtering is a guard, not permission to silently discard failed simulations: validation checks all saved means are finite and length B. For Cauchy these finite-vector diagnostics do not estimate existing population moments. Their population targets are undefined; no standardization by a theoretical Cauchy SD is used.

Reproduction and sources

Configuration: confirmed student seed 7616; fixed sensitivity seed 7617; B=10,000. R 4.6.1, Quarto 1.10.18; dependency versions in `renv.lock`. The saved diagnostic tables, judgment CSV and plotted vectors are distinct: numerical results are generated; choosing a practical boundary remains a judgment.

From the repository root, restore and regenerate:

```
Rscript -e 'renv::restore(prompt = FALSE)'
Rscript scripts/smoke_test.R
Rscript scripts/run_project.R
Rscript scripts/validate_results.R
Rscript scripts/render_summary.R
quarto render
```

Validation covers theoretical moments/SE, nonexistence of Cauchy references, varying basketball probabilities, preserved failure recurrence, both seeds' vector/CSV agreement, B and finite means, unique diagnostic rows and evidence-linked final selections. A fresh-directory rerun on the same R version reproduced primary/sensitivity diagnostics, calibration and final summary byte-for-byte. All eight individual Quarto reports remain saved and rendered; this PDF contains their grading evidence without requiring those files to be opened.

Sources read and preserved:

1. *CLT Project 1 Student Guidance* (`clt_project_1_student_guidance.qmd`): updated five-required/three-extra scope and presentation requirements.
2. *Math167R Class 06 Project 1: CLT Monte Carlo*: the three questions, broad-to-fine investigation, evidence, seed sensitivity and AI iteration.
3. *Math167R Class 05 Central Limit Theorem*: iid finite-variance theory, means and sampling distributions.
4. *Math167R Class 06 Statistical Distributions*: population characteristics and R distribution functions.
5. *Dependent Failure Process*: exponential rate recurrence and parameters.
6. *Non-Identical Basketball Shots*: independent varying-probability construction, adapted to variable n as specified.
7. *Custom Age-at-Death Distribution Monte Carlo*: synthetic two-component discrete weights and rdeath construction.

Original instructor files are preserved unchanged in `references/`. Lesson exercises unrelated to the updated project population list are background rather than extra submission tasks. Expectations in this PDF are explicitly retrospective because a contemporaneous pre-run prediction was not recorded; it would be misleading to manufacture one now.

Repository: github.com/babaJaan01/Monte_Carlo_CLT. It contains the eight `.qmd` investigations, reusable R functions, generated results, full AI log and downloadable PDF. The scientific argument and required evidence are included here; the repository is for reproduction and optional deeper inspection.